WORKED SOLUTION

STEP I 2006, Question 8

Pure • Vectors in Three Dimensions

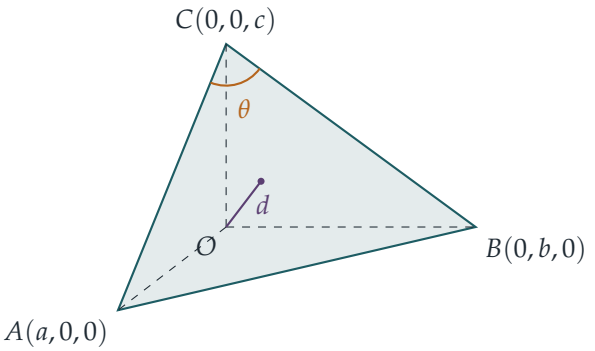
STEP I 2006, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q8

The key idea

The tetrahedron $OABC$ has three mutually perpendicular edges at O, so its volume is easy if we use OAB as the base. The same solid can also be viewed with ABC as the base, and then the height is exactly the distance d we want. Working out the volume in two ways gives an equation for d, provided we know the area of ABC. That area comes from the angle at C, which the scalar product gives us.



Solution

Part (i)

Take triangle OAB, which lies in the plane $z = 0$, as the base. Since OA and OB lie along the x- and y-axes, the angle AOB is a right angle, so

$$\text{area of } OAB = \tfrac{1}{2}ab.$$

The height is the distance of C from the plane $z = 0$, which is c, because OC lies along the z-axis and so is perpendicular to that plane. Hence

$$V = \tfrac{1}{3} \times \tfrac{1}{2}ab \times c = \frac{abc}{6}.$$

Part (ii)

The angle ACB is the angle between the vectors $\overrightarrow{CA}$ and $\overrightarrow{CB}$:

$$\overrightarrow{CA} = \begin{pmatrix} a \\ 0 \\ -c \end{pmatrix}, \qquad \overrightarrow{CB} = \begin{pmatrix} 0 \\ b \\ -c \end{pmatrix}.$$

Their scalar product is $0 + 0 + c^2 = c^2$, and their lengths are $\sqrt{a^2 + c^2}$ and $\sqrt{b^2 + c^2}$. So

$$\cos\theta = \frac{\overrightarrow{CA} \cdot \overrightarrow{CB}}{|\overrightarrow{CA}|\,|\overrightarrow{CB}|} = \frac{c^2}{\sqrt{(a^2 + c^2)(b^2 + c^2)}},$$

which is the required result.

For the area we need $\sin\theta$. Since $0 < \theta < \pi$, $\sin\theta > 0$, so

$$\begin{aligned} \sin\theta = \sqrt{1 - \cos^2\theta} &= \sqrt{\frac{(a^2 + c^2)(b^2 + c^2) - c^4}{(a^2 + c^2)(b^2 + c^2)}} \\ &= \frac{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}{\sqrt{(a^2 + c^2)(b^2 + c^2)}}, \end{aligned}$$

because the c^4 terms cancel when we expand $(a^2 + c^2)(b^2 + c^2)$.

> **Ask yourself** Which formula for the area of a triangle fits the information we now have?

We know two sides, CA and CB, and the angle θ between them, so we use $\frac{1}{2}\,CA \cdot CB \sin\theta$. The awkward square roots cancel:

$$\text{area of } ABC = \tfrac{1}{2}\sqrt{(a^2 + c^2)(b^2 + c^2)} \cdot \frac{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}{\sqrt{(a^2 + c^2)(b^2 + c^2)}} = \tfrac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2}.$$

Pleasingly, the answer is symmetric in a, b and c, as it must be: the triangle ABC doesn't care which vertex we called C.

The distance d

Now regard ABC as the base of the same tetrahedron. Its height is the perpendicular distance from the opposite vertex O to the plane of ABC, which is d. Since the volume doesn't depend on which face we call the base, part (i) and part (ii) give

$$\frac{abc}{6} = \tfrac{1}{3} \times \tfrac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2} \times d, \qquad \text{so} \qquad d = \frac{abc}{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}.$$

Squaring and inverting (everything is positive),

$$\frac{1}{d^2} = \frac{a^2b^2 + b^2c^2 + c^2a^2}{a^2b^2c^2} = \frac{1}{c^2} + \frac{1}{a^2} + \frac{1}{b^2},$$

dividing each term of the numerator by $a^2b^2c^2$. This is the required result.

Checks

- Take $a = b = c = 1$. Then ABC is equilateral with side $\sqrt{2}$, of area $\frac{\sqrt{3}}{4} \cdot 2 = \frac{\sqrt{3}}{2}$, which matches $\frac{1}{2}\sqrt{3}$. The plane is $x + y + z = 1$, at distance $1/\sqrt{3}$ from O, and indeed $1/d^2 = 3$.
- With $a = 1$, $b = c = 2$ the formulae give volume $\frac{2}{3}$, area $\sqrt{6} \approx 2.449$ and $d = 2/\sqrt{6} \approx 0.8165$, so $1/d^2 = \frac{3}{2} = 1 + \frac{1}{4} + \frac{1}{4}$.
- Numerically, for 10 000 random (a, b, c), the volume (by the scalar triple product), the area (by the cross product) and d (by locating the foot of the perpendicular directly) all agreed with the formulae to within 10^{-12}.

Common slips

- Using $\overrightarrow{AC}$ with $\overrightarrow{CB}$: mixing directions flips the sign of the scalar product and gives the supplement of θ.
- Forgetting that the height for base ABC is the perpendicular from O to the plane, not the length of any edge.

Beyond STEP

The plane through A, B and C is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, with normal $(\frac{1}{a}, \frac{1}{b}, \frac{1}{c})$, and the general point-to-plane distance formula gives d in one line. The area result is a three-dimensional Pythagoras theorem (de Gua's theorem): the squared area of ABC equals the sum of the squared areas of the three right-angled faces at O, since $\frac{1}{4}(a^2b^2 + b^2c^2 + c^2a^2)$ is exactly that sum. Both ideas come from the cross product and determinants, which you meet early in a first linear algebra course; Strang's *Introduction to Linear Algebra* is a good place to see them.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.